Daniel A. Delgado, PhD

Product and Interaction Designer | Spatial Computing | Human-Centered Design
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Independent Product Design Projects

Galaxy Viz – 3D Data Visualization 2026–Present
Independent Project | Product Concept, Interaction Design, and Prototyping

- Conceived and developed an interactive 3D interface for exploring academic research as a navigable visual environment, using spatial layout, color, scale, and connected networks to communicate complex relationships.
- Designed search, timeline controls, navigation, and visual encodings; developed features that surface disconnected research topics and conflicting findings.
- Built and deployed the experience using Three.js, WebGL, TypeScript, and a data-processing pipeline.

iCarus – Skinnable macOS Media Player 2026–Present
Independent Project | Product Design, UI Architecture, and Development

- Designed and developed a native media-player experience in SwiftUI, integrating audio/video playback, media downloads, and customizable interface layouts.
- Engineered a modular skin system supporting custom visual styles and Windows Media Player skin imports, including JavaScript-driven animations and interactions.

University of Florida | Spatial Computing and Interaction Design

Enhancing Collaborator Communication in Extended Reality Aug. 2024–Jul. 2025
XR Researcher and Software Developer | Human-Computer Interaction

- Led mixed-methods research evaluating shared-gaze visualization concepts for collaborative industrial AR, translating findings into design recommendations for iterative prototype development.
- Designed and conducted three controlled user studies (n=60+) using usability testing, behavioral observation, and quantitative analysis to assess collaboration, social presence, and task performance.
- Developed AR prototypes integrating HoloLens 2 eye tracking, real-time gaze visualization, spatial interfaces, and interactive feedback; implemented multiplayer interactions and cross-device synchronization using Unity, C#, Photon, and Azure Spatial Anchors.

ENKIx – Enabling Knowledgeable Task Guidance in the Extremes Jun. 2021–Jul. 2024
User Interface Designer

- Developed an augmented-reality prototype for context-aware task guidance using Unity and C#/.NET; implemented computer-vision object detection with ArUco markers.
- Designed interactive interfaces and collaborated with multidisciplinary researchers and engineers to improve task guidance.

EMG-Based Prosthetic Simulation in VR Jun. 2020–Jun. 2023
User Researcher and Software Developer

- Compared cognitive workload and usability of EMG-based prosthetic training in Unity VR versus physical devices through structured user evaluations.
- Designed study protocols, recruited participants, conducted usability evaluations, and analyzed qualitative and quantitative data.
- Developed interactive VR simulations integrating Delsys EMG sensors, Leap Motion hand tracking, and machine-learning models for real-time gesture classification and prosthetic control.

Education

Ph.D., Computer Science 2025
University of Florida, Gainesville, FL
Dissertation: *Enhancing Collaborator Communication Through Shared-Gaze Visualizations in Augmented Reality*

Design, Research, and Technical Skills

Product and Interaction Design: Interaction design, UI architecture, user-centered design, Figma, prototyping, usability testing, human factors
UX Research and Analytics: Mixed-methods research, user interviews, contextual inquiry, survey design, experimental design, behavioral analysis, Qualtrics, R, Python, MATLAB, ANOVA, regression
Spatial and Immersive Experiences: AR/VR, Unity, HoloLens 2, eye tracking, hand tracking, computer vision, wearable sensing
Visual and Interface Development: SwiftUI, Three.js, WebGL, GLSL, 3D visualization, responsive interfaces, C#/.NET, JavaScript, TypeScript
AI and Development Tools: TensorFlow, Git, Claude, AI-assisted research and development